



SMART AGRICULTURAL EXTENSION RETRIEVAL SYSTEM

A hybrid retrieval-augmented generation (RAG) pipeline for agronomic advisory search

Team Tsavo | TRI AI Cohort 10 Benchmark Submission

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Public Leaderboard: **0.86972 MRR@10** · Rank 9

Bridging the Agronomic Advisory Gap for Extension Workers



Field Diagnostic Urgency

Extension officers in Sub-Saharan Africa need rapid, accurate diagnostic advice for plant diseases, pests, and soil health during field visits.



Complex Agronomic Data

Unstructured factsheets are hard to search: specialised chemical codes (NPK ratios, pesticide active ingredients) and varying crop terminology confuse naive search.



High Misdiagnosis Cost

Incorrect recommendations lead to crop failure, wasted treatment expenses, and yield reduction for smallholder farmers.

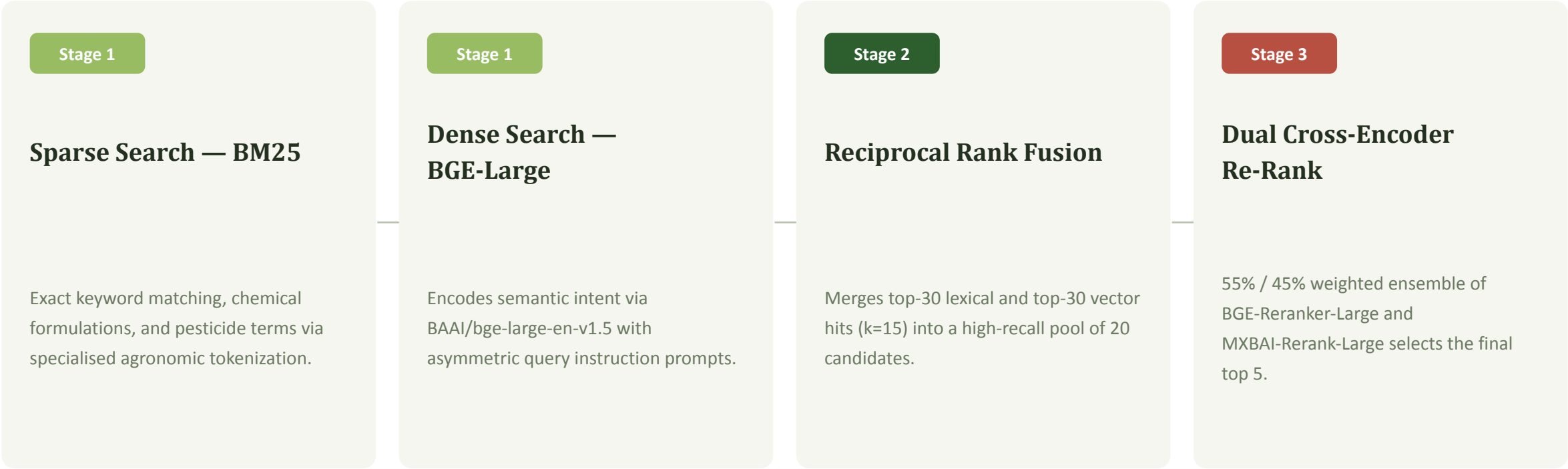
TARGET OBJECTIVE

Deliver a high-precision retrieval pipeline yielding grounded, zero-hallucination advice in the top-5 search ranks.

Evaluation metrics

nDCG@5 · MRR@10

Two-Stage Hybrid Retrieval & Dual Re-Ranking Engine



Maximizing Vector Attention via Unified Schema Enrichment

METADATA FUSION SCHEMA

```
Title: [Title] | Crop: [Crop] | Source:
[Source] | Text: [Text]
```

- **Token Preservation Rules**

Custom regex tokenization preserves hyphenated NPK ratios (e.g., 15-15-15) and alphanumeric product codes.

- **HyDE Removal Strategy**

Stripped synthetic query-expansion preambles after evaluation showed synthetic noise reduced overall MRR.

- **Corpus Preprocessing**

Imputed missing crop values to 'general' and normalized whitespace across all 695 document factsheets.



695

document factsheets indexed

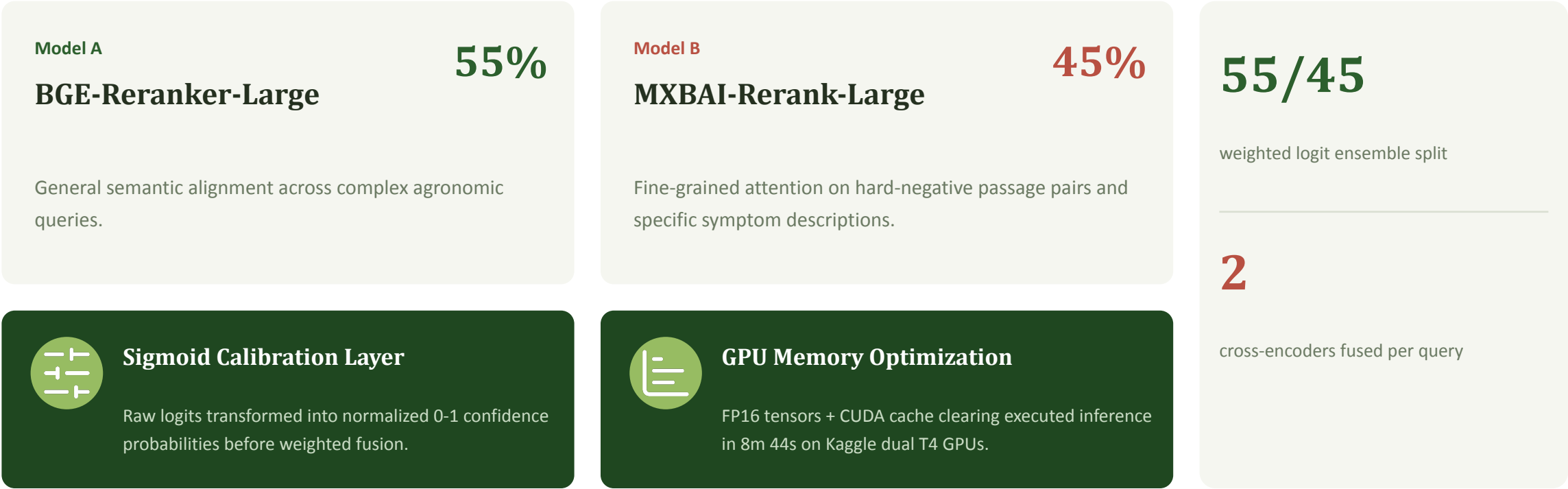
Fields fused per document

Title · Crop · Source · Body Text

Normalization

Missing crop - general; whitespace cleaned

Calibrated Multi-Model Re-Ranking Strategy



PERFORMANCE METRICS

Kaggle Benchmark Results & Execution Metrics

PUBLIC LEADERBOARD SCORE

0.86972

MRR@10 · Rank 9 of 25 Teams

Notebook `notebook0f717eabbaOgeAI` (v1, peak score)

EXECUTION ENVIRONMENT

Kaggle Dual Tesla T4 GPUs

TOTAL RUNTIME

8 min 44 sec (524.2s)

EVALUATION SET

200 queries - 1,000 rows

SUBMISSION FORMAT

QueryId, DocumentId
(long)

Translating Search Benchmarks into Field Advisory Value



Accelerated Field Diagnosis

Fast retrieval (<3s per query) lets extension agents deliver immediate advisory information during farm visits.



Yield Protection

High precision in top ranks prevents confusing fungal infections with nutrient deficiencies, saving crops and input costs.



Grounded Advice

RAG design ensures retrieved answers originate from vetted extension manuals (FAO, CGIAR, Plantwise).



Transparency

Metadata tags (Crop, Source, Authority) accompany retrieved text blocks to preserve regional context.

Production Readiness & Future Scale



Modular Repository

Fully documented codebase published at github.com/ogeitunu/C10-team-tsavo, adhering to modular Python standards.



Edge & WhatsApp Deployment

Quantizing re-rankers for low-bandwidth WhatsApp Business API and offline mobile search.



Multilingual Integration

Translation layers for localized languages (Swahili, Hausa, Yoruba) for direct farmer access.



Thank You!

Ready for Demo Day evaluation and Q&A.

Team Tsavo — TRI AI Cohort 10